SEPTEMBER 2024

JAPAN RATES PRIMER

An overview of Japan's rates market

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Stephen SPRATT historical MAD2MAR recommendations over past 12 months

SOCIETE GENERALE
Cross Asset Research

BOJ RATES

- Objective: BoJ sets policy to achieve the 'price stability target' which it defines as 2% yoy in CPI
- BoJ sets interest rate target for uncollateralized call rates (similar to Fed funds), where financials lend/borrow overnight. <u>Current target is around 0.25%</u>
- Banking system cash supply/demand is forecast each day by BoJ. It then adds/removes liquidity via repo operations to maintain the level of rates
- Add liquidity = 'fund supplying' operations = BoJ lends cash vs collateral
- Removing liquidity = 'funds absorbing' operations = BoJ lend collateral vs cash
- Changes in interbank cash demand often happen during large volume transactions, such as tax payment dates, large government bond issuance, etc.

日本銀行 BANK OF JAPAN

Results for Sep. 3 (Tue)

Bank of Japan
Financial Markets Department

(100 million yen, percent per annum)

Instrument	Amounts of Competitive Bid*	Amounts of Successful Bid*	Pro-rata Rate/Yield Spread** /Price Spread***	Non-pro-rata Rate/Yield Spread** /Price Spread***	Average Successful Bid Rate/Yield Spread** /Price Spread***	Allocation on a Pro-rata Basis Rate
US Dollar Funds-Supplying Operations against Pooled Collateral (Starting on Sep. 5)	1	1				
Securities lending facility (Sales of JGSs under repurchase agreements) /offered in the morning/same-day-start operation****	16,827	16,827		-0.050	-0.050	
Funds-Supplying Operations against Pooled Collateral (at All Offices) (Fixed rate) (Starting on Sep. 4)	11,806	8,004				67.8

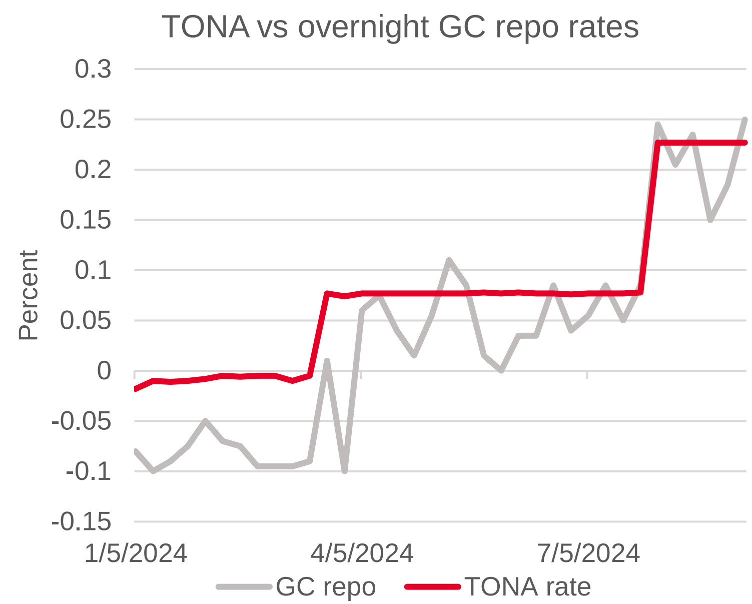
Source: BOJ https://www3.boj.or.jp/market/en/stat/ba240903.htm

BOJ RATES

- BoJ publishes weighted average rates of all unsecured overnight transactions for the prior day at 10am local time.
- This is known as Tokyo Overnight Average Rates (TONA) and is the reference rate for Japan swaps

Other parts of funding markets

- GC repo rate: collateralised borrowing against JGBs. Rate loosely follows TONA and the collateral/cash balance in the system
- GC repo can trade above (below) call rates when too much (not enough) collateral is available
- Domestic banks can arbitrage transactions between the two markets, borrowing/lending in TONA to invest/borrow in GC at a profit
- This keeps the two in balance. Many overseas participants do not have access to TONA



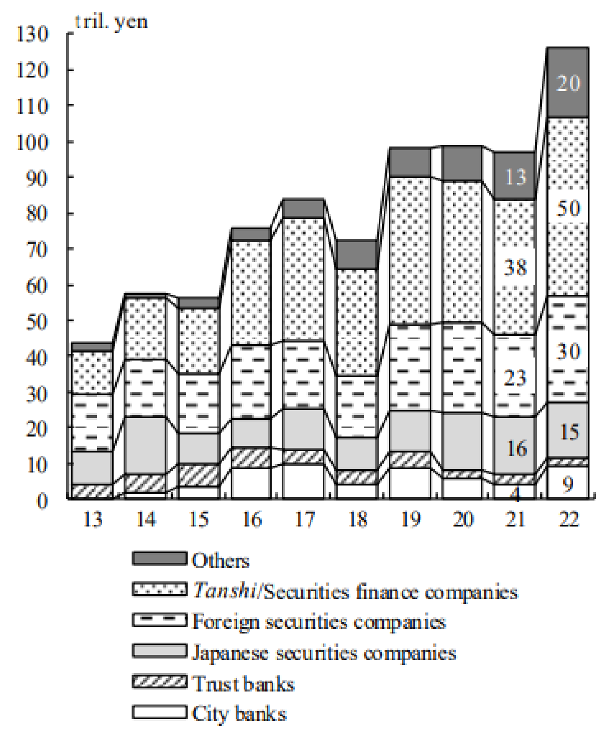
Source: SG Cross Asset Research, Bloomberg

GC REPO MARKET

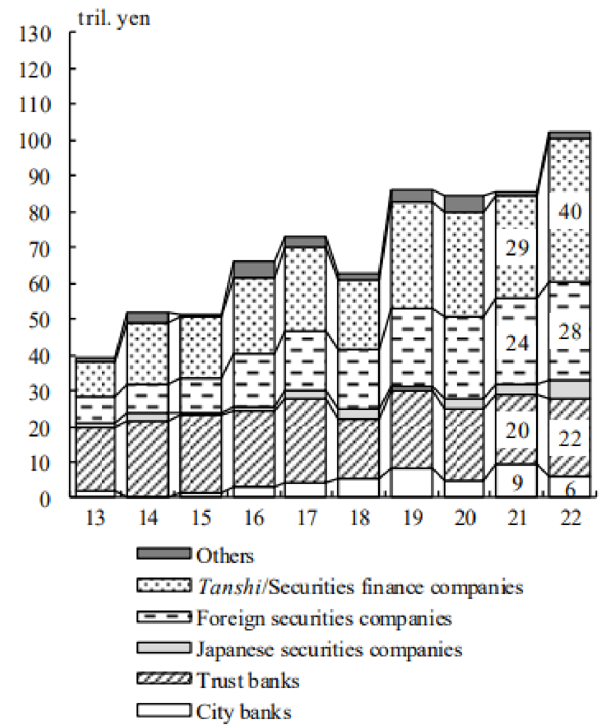
- BoJ data shows that domestic and foreign securities dealers typically run sizable net GC repo positions (cash borrowing)

- This is used to finance inventory needed for GC market-making and trading in the special collateral (SC) market
- BoJ's bond buying has seen SC trading become more common
- These borrowers face cash-rich Trust banks who are the cash lenders

Cash Borrowing Side



Cash Lending Side

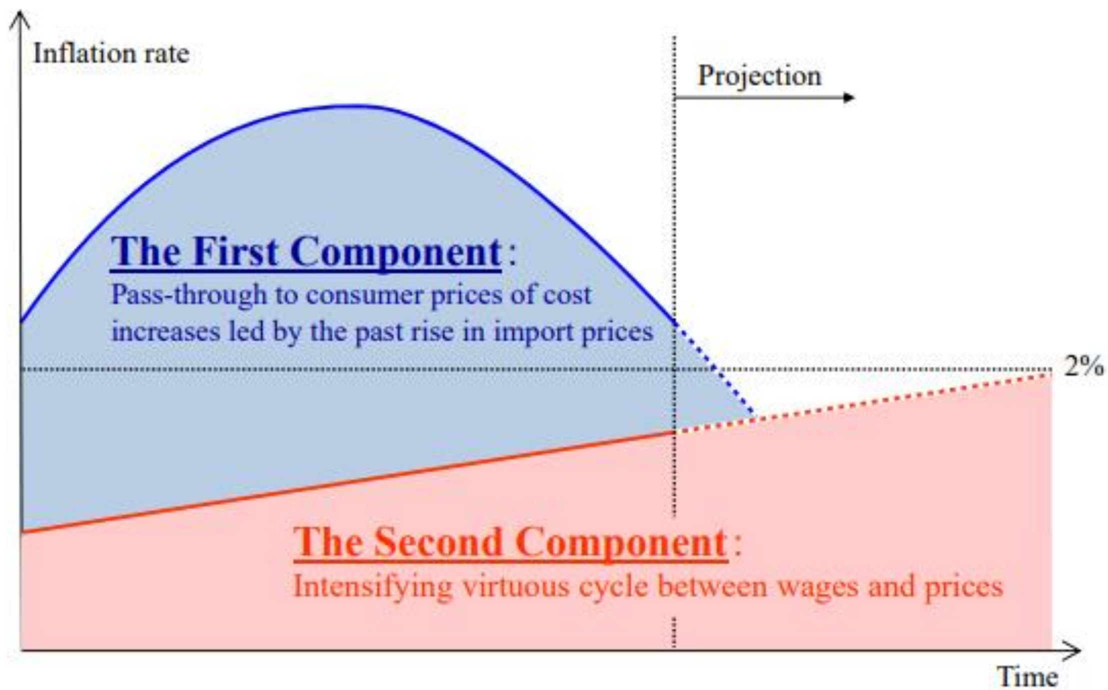


Source: BoJ - Trends in the Money Market in Japan, https://www.boj.or.jp/en/paym/market/data/market2211.pdf

JAPAN MOVES FROM DEFLATION TO REFLATION

FX did the heavy lifting: Last three years -25% vs USD

- 'Cost push' inflation via FX channel/import prices morphed into local wages and price pressured
- Governor Ueda laid this out in a speech in May



* BoJ: Virtuous Cycle between Wages and Prices and the Bank of Japan's Monetary Policy (boj.or.jp)

BOJ RATES – A SHORT HISTORY

Interest rates	Bond buying	Forward guidance
Jan. 2016: BoJ moves to tiered IOER regime* The tiers: **0.1%** Basic rate (Benchmark balance - reserves) **0%** Macro add (Benchmark balances * ratios + lending programmes) **-0.1%** Policy-rate balance (current account balance - macro & basic balance) Objective: realize negative rate impact without being too punitive on financial institutions' earnings	**YCC: 10y target set at "around 0%"** How upper band evolved: • July 2018 0.1% • March 2021: 0.25% • December 2022: 0.5% • July 2023: 1% • October 2023: 1% a "*reference*" in operations, not hard cap*** BoJ slowly tweaks purchase sizes to reduce supply/demand imbalances	**Persistent easing** BoJ to maintain QQE with YCC for as long as necessary to achieve inflation target in a stable manner. Expanding the monetary base until yoy CPI exceeds 2% and stays above target in a stable manner
March 2024: hikes rates to 0.1%, scraps tiering** • Applies 0.1% rate across current account balances (excluding required reserves) • Uses this to guide uncollateralised overnight call rate to around 0 – 0.1%	**Scraps YCC target** • BoJ says it will continue to buy bonds at current pace of around JPY6tn/month	BoJ says inflation-overshooting commitment re: monetary base had been fulfilled. "accommodative financial conditions will be maintained for the time being"
July 2024: BoJ raises call rate to around 0.25% • Applies 0.25% rate across current account balances • Guides uncollateralised call rate to around 0.25%	**Plans to reduce bond buys** • Will cut by JPY400bn/qtr to "about" JPY3tn/month in 2026 • Allow markets to set long-term yield ranges	If outlook evolves as forecast, BoJ will "continue to raise the policy interest rate and adjust the degree of monetary accommodation"

**BoJ: https://www.boj.or.jp/en/research/brp/ron_2022/data/ron220527a.pd*
*** BoJ: https://www.boj.or.jp/en/mopo/mpmdeci/mpr_2024/k240123a.pdff*
**** Reuters: https://www.reuters.com/markets/asia/how-does-japans-yield-curve-control-work-2023-01-16/*

BOJ BOND BUYING

- BoJ's first QE was in March 2001-2006. It then bought assets October 2010. QQE began in April 2013. Formal tapering began in August, now buying JPY5.3tn a month
- Regular QE buying is known as Rinban operations

How it works

- BoJ announces buying schedule each quarter
- Sizes confirmed on the day of operations at 10:10 local time, with cutoff for offers at 11:40. Results released at around 12:00
- Banks & financials with BoJ accounts put in offers to sell securities. Investors without access can do so through banks
- BoJ fills offers starting at the highest yield in a competitive auction process

BoJ Aug-Sept. bond-buying plan

	残存期間 Residual maturity	1回当たりオファー金額[2] Purchase size per auction （単位：億円） (100 million yen)	月間オファー金額 Monthly purchase size （単位：億円） (100 million yen)	オファー日程[3] （8-9月分） Scheduled dates (August-September)
利付国債[1] JGBs with coupons	1年以下 Up to 1 year	1, 500	1, 500	8/2 9/4
	1年超3年以下 More than 1 year and up to 3 years	3, 500	14, 000	8/2、7、15、21 9/4、11、18、25
	3年超5年以下 More than 3 years and up to 5 years	3, 750	15, 000	8/2、7、21、28 9/4、11、18、25
	5年超10年以下 More than 5 years and up to 10 years	4, 000	16, 000	8/2、15、21、28 9/6、11、18、25
	10年超25年以下 More than 10 years and up to 25 years	1, 500	4, 500	8/7、15、28 9/6、18、25
	25年超 More than 25 years	750	1, 500	8/15、28 9/6、11
物価連動債 Inflation-indexed bonds		600	600	8/28 9/25
			計　53, 100	

Source: BoJ 長期国債買入れ（利回り・価格入札方式）の予定（2024年8～9月）［日程更新］(boj.or.jp)

BOJ BOND BUYING

What to watch in Rinbans

- Use the results to gauge investor willingness to sell:
- Bid/cover ratio (although technically offer/cover ratio) = High (low) shows strong (weak) willingness to sell JGBs in the specified sector
- Tail = spread between yield level the BoJ buys, and the reference rate (prior day close) is published as the average spread. It is important to consider yield moves between the prior close and the operation cut-off
  - If BoJ buys bonds at much higher yields/lower price = wider tail = stronger appetite to sell (and vice versa)

Other QE-buying operations

- Fixed rate purchases of unlimited size known as ‘Sashine’ operations
- First in effect when YCC introduced in 2016 (YCC now scrapped)
- Focused on 10y to put a hard cap on yields. Rarely used, mostly to facilitate yield targets under YCC or when prices drop sharply

THE PROBLEM WITH QE FOR SO LONG…

- BoJ owns around 50% of the bond market. For some specific issues it is over 90%
- This leads to periodic shortages of bonds and specials in repo (SC). BoJ eases some of this by meeting requests from dealers via their securities lending programme.*
- BoJ % holdings of issues across major maturities:

2y

Bond	Maturity	BoJ %	Years to maturity
JB347	6/20/2027	88.7%	2.8
JB345	12/20/2026	87.1%	2.3
JB346	3/20/2027	87.0%	2.5
JB344	9/20/2026	86.2%	2.0
JS149	9/20/2026	73.8%	2.0
JS153	6/20/2027	72.9%	2.8
JS150	12/20/2026	70.6%	2.3
JS151	3/20/2027	68.8%	2.5
JL95	6/20/2027	59.7%	2.8
JS152	3/20/2027	59.2%	2.5
JL91	9/20/2026	56.2%	2.0
JL92	12/20/2026	50.6%	2.3
JL93	3/20/2027	50.3%	2.5
JL96	6/20/2027	50.0%	2.8
JL90	9/20/2026	46.0%	2.0
JL94	3/20/2027	45.0%	2.5

5y

Bond	Maturity	BoJ %	Years to maturity
JB356	9/20/2029	88.8%	5.0
JB359	6/20/2030	88.9%	5.8
JB357	12/20/2029	87.8%	5.3
JB358	3/20/2030	85.5%	5.5
JL113	9/20/2029	73.1%	5.0
JL117	3/20/2030	69.6%	5.5
JL114	12/20/2029	65.2%	5.3
JL116	3/20/2030	60.5%	5.5
JL119	6/20/2030	57.0%	5.8
JL120	6/20/2030	54.7%	5.8
JL115	12/20/2029	53.5%	5.3
JX2	2/20/2030	51.1%	5.4
JL118	6/20/2030	48.7%	5.8
JX3	5/20/2030	39.6%	5.7
JX1	9/20/2029	7.6%	5.0

7y

Bond	Maturity	BoJ %	Years to maturity
JB366	3/20/2032	95.1%	7.5
JB367	6/20/2032	89.5%	7.8
JB365	12/20/2031	83.6%	7.3
JB364	9/20/2031	82.7%	7.0
JL137	6/20/2032	55.7%	7.8
JL138	6/20/2032	55.6%	7.8
JX6	11/20/2031	51.4%	7.2
JL139	6/20/2032	50.0%	7.8
JL136	3/20/2032	44.1%	7.5
JL135	3/20/2032	43.1%	7.5
JL131	9/20/2031	41.4%	7.0
JL132	12/20/2031	36.7%	7.3
JL133	12/20/2031	36.2%	7.3
JX7	5/20/2032	35.2%	7.7
JL130	9/20/2031	35.0%	7.0
JL134	3/20/2032	33.3%	7.5

10y

Bond	Maturity	BoJ %	Years to maturity
JL153	6/20/2035	59.4%	10.8
JL151	12/20/2034	57.3%	10.3
JL150	9/20/2034	52.5%	10.0
JL152	3/20/2035	51.7%	10.5
JX16	9/20/2034	31.9%	10.0
JX17	12/20/2034	29.4%	10.3
JX19	6/20/2035	24.7%	10.8
JX18	3/20/2035	13.6%	10.5

30y

Bond	Maturity	BoJ %	Years to maturity
JU12	3/20/2059	42.8%	34.5
JU11	3/20/2058	39.1%	33.5
JU13	3/20/2060	38.4%	35.5
JU14	3/20/2061	30.6%	36.5
JU10	3/20/2057	26.7%	32.5
JU8	3/20/2055	12.3%	30.5
JU15	3/20/2062	11.9%	37.5
JU16	3/20/2063	7.0%	38.5
JU9	3/20/2056	4.3%	31.5
JU17	3/20/2064	4.1%	39.5

Source: SG Cross Asset Research; BoJ, Bloomberg

*BoJ https://www.boj.or.jp/en/mopo/measures/mkt_ope/ope_b/opetori11.htm-

MOF SUPPLY VS BOJ BOND BUYING

BoJ July meeting: will reduce bond buying pace by JPY400bn/quarter to "about" JPY3tn/month by 2026. <u>BoJ QE declines, net supply rises</u>

- The skew of BoJ buying across maturities has left supply/demand tightest in below 10y maturities. <u>We expect future QE reductions will be focused here</u>

- Central bank bond buying has now slowed below the pace that existing holdings are rolling off
- BoJ balance sheet is now slowly shrinking (QT) and is projected to fall by 7-8% over next three years (equiv. $315bn)*
- Slow pace of QT in theory should put some additional upward pressure on yields over time. We expect a tightening impact on swap spreads

Rinbans vs JGB supply

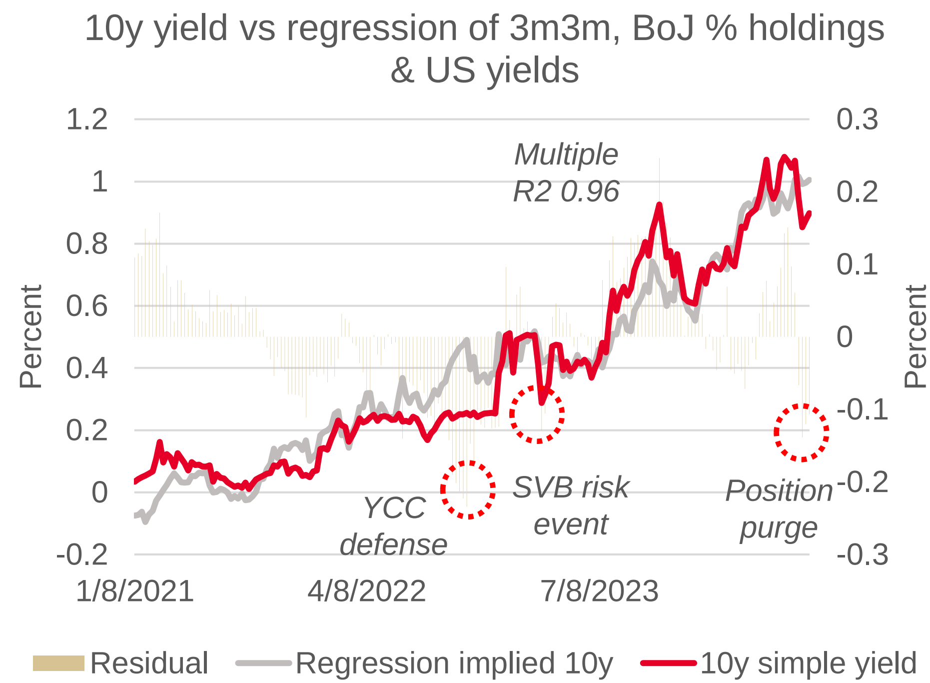
Buckets	Sept. Rinbans (JPY bns)	Sept. Supply (JPY bns)	%
1~3yr	1400	2850	49
3~5yr	1500	2550	59
5~10yr	1600	2300	62
10-25yr	450	1350	33
25yr~	750	1600	47

Source: SG Cross Asset Research; Japan's MoF,BoJ

* *July BoJ: <u>Decisions at the July 2024 Monetary Policy Meeting (boj.or.jp)</u>*

Make JGBs great again

- Japan has the second-largest debt market in the world but is dominated by domestic investors
- At end-2023: foreign investors <7% of JGBs but around 66% of T-bills* due to favourable basis (more below)
- After years of poor volumes, suppressed volatility due to weak inflation and policy responses (YCC and QE) it is now coming back to life
- 10y JGB yields are heavily influenced by short-end rates, BoJ ownership of bonds and US yields



Source: SG Cross Asset Research; Bloomberg
*Regression = -1.57 + 1.68 * JPY 3m3m + 0.144 * US 10y yield + 2.92 * BoJ % JGB ownership*

** MOF: https://www.mof.go.jp/english/policy/jgbs/publication/newsletter/jgb2024_06e.pdf*

JAPAN BONDS: TRADING IT

Investors can look at outright yields, curve spreads, flys

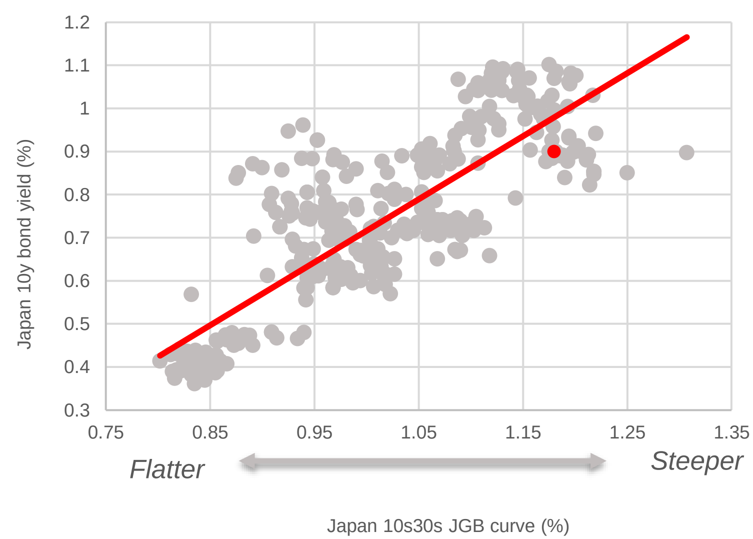
- Model independent: Z-scores to gauge how cheap/rich a spread is versus a normal distribution
  - Other standardised residuals for curves and flys, versus yield levels (e.g. 10s30s vs level of 10y rates)
- Model dependent: Spline-based or parametric models (Nelson Siegel) for curves, or Principal Component Analysis (PCA) to distil rate drivers into level, curve and slope
- Macro models: use market-based measures, economic variables and central bank rate inputs to generate 'fair value' estimates

JPY OIS curve spreads

Rates	Z-score			Residual vs yield levels
JPY OIS	3m	6m	12m	3m
OIS flys				
1s2s3s	0.2	0.2	0.9	-0.3
2s5s7s	-1.0	-1.5	-1.6	1.0
2s5s10s	-0.9	-0.9	-1.2	0.9
2s5s20s	-1.2	-1.0	-1.3	1.2
2s7s10s	-1.1	-1.8	-1.8	1.1
2s7s15s	-1.3	-2.0	-1.9	1.3
2s7s20s	-1.4	-2.1	-1.9	1.4
2s7s30s	-1.5	-2.1	-1.8	1.6
2s10s20s	-1.2	-1.9	-2.0	1.2
2s10s30s	-1.4	-2.1	-1.9	1.4

Source: SG Cross Asset Research, Bloomberg

Curve appears steep for level of yields



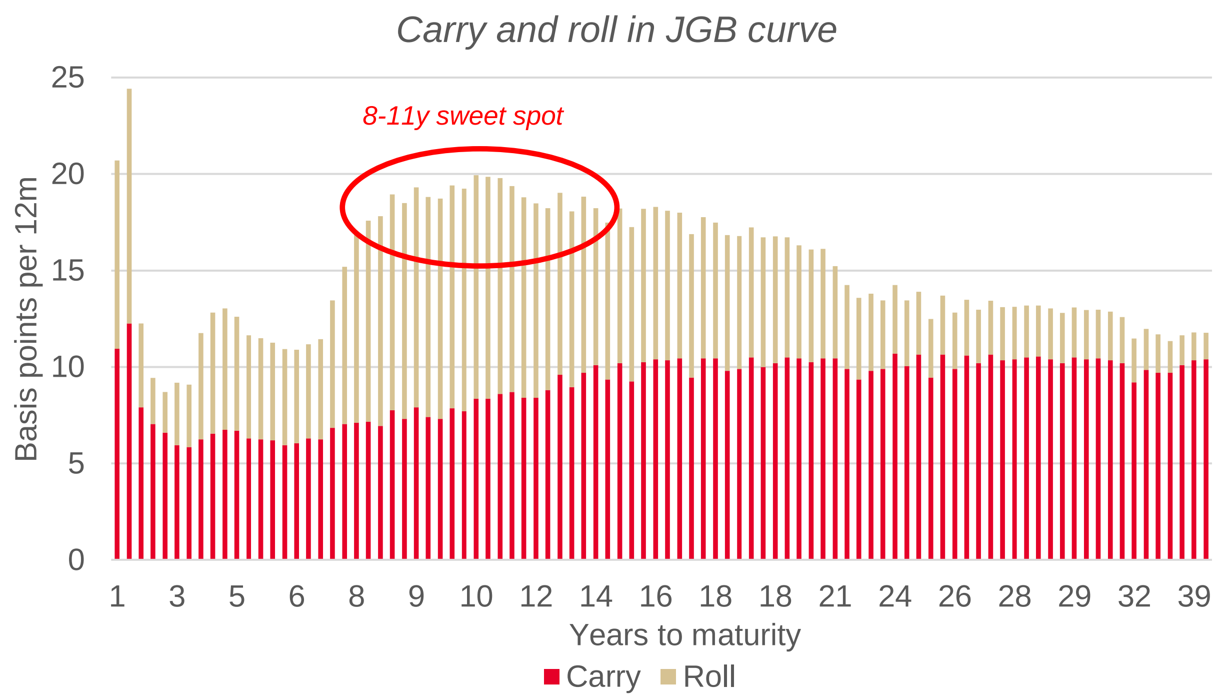
Source: SG Cross Asset Research, Bloomberg

Carry and rolldown (*aka carry & slide*)

- Investors picking the best points to own, which offer the highest combination of:

- Carry (collecting the semi-annual coupon payments)
- Rolldown (the yield "slide" assuming the curve remains unchanged between today and a date in the future.
- E.g. 1y rolldown on a 10y would be 10y yield today - 9y yield today)

BBG: {GOVY <GO>}



Source: SG Cross Asset Research, Bloomberg

JAPAN AUCTION CYCLE

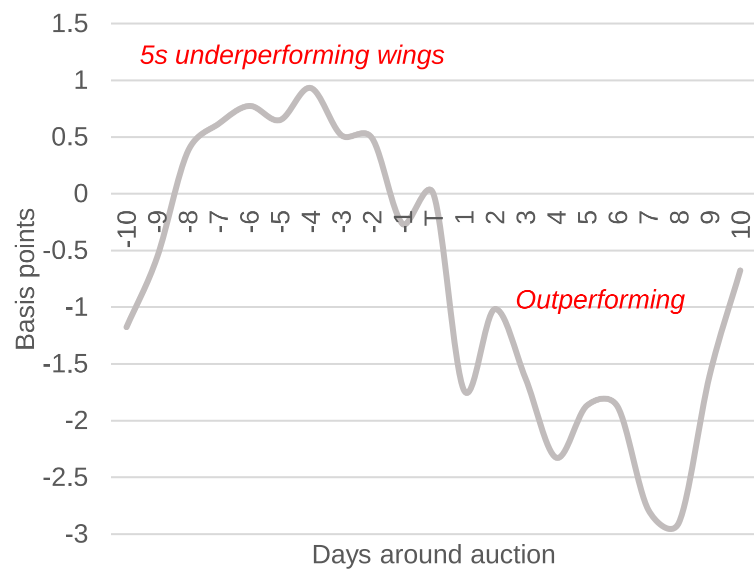
- Japan sells 3m bills weekly and 6m, 1y monthly; 2y, 5y (medium term) 10y (long), 20y, 30y bonds (super longs) are sold 1x month; 40y bonds (super longs), 10y linkers are sold 6x year
- Most auctions are price-competitive (conventional style). Highest bid filled first, then second highest etc. until total size is allocated
- Exceptions: 40y & linkers are 'dutch-auction' style (i.e. one yield/price for all bidders)
- Regular "liquidity enhancement" auctions are held to issue older bonds lacking liquidity
- Japan has a primary dealer system. There are 19 dealers (including Societe Generale Securities Japan Limited). Together, they are required to bid for 100% of each auction*
- How to determine how an auction goes? Pricing gives best indication
    - Bloomberg publishes a poll of the expected lowest price for the sale, which can be compared to actual price
    - The 'tail', the spread between average price and lowest price paid, shows how aggressively investors were willing to bid for the bonds
    - Bid/cover ratios give a broader gauge of demand

* MOF: JGB Market Special Participants Scheme : Ministry of Finance (mof.go.jp)

JAPAN AUCTION CYCLE

- Auctions tend to build in concessions (in latin “concedere” = “to give way”)
- Dealers together need to bid 100%. If they do not want to add duration, they can offset this by getting short bonds prior
- Some investors prefer to own more recent bonds as they are typically more liquid
- Bonds tend to lag peers into sales as “concession” builds, when dealers and investors make room for supply
- This tends to be unwound after auctions but can depend on the auction results
- Playing supply dynamics around auctions vs Rinbans has been a common RV strategy

Past 12m: On average 5y lags on 2s5s10s into auction

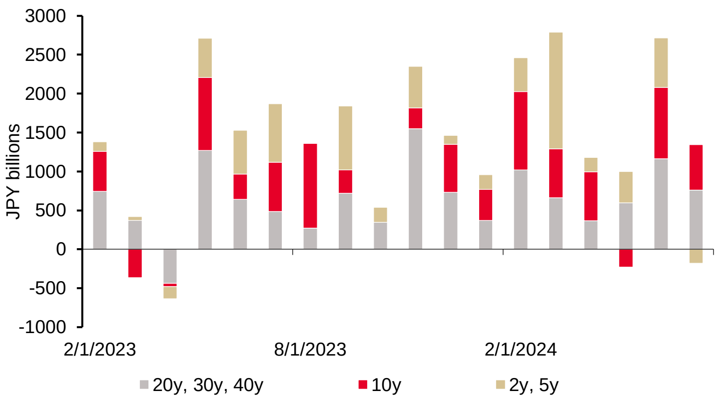


Source: Bloomberg, SG Cross-Asset Research

JAPAN BONDS: WHO BUYS WHAT

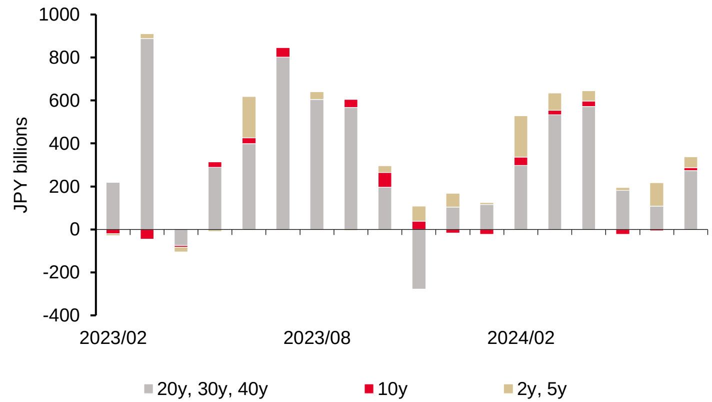
Flow to know: Monthly JSDA release

Trust banks (proxy for pensions) buy across the curve, skewed to 10y+



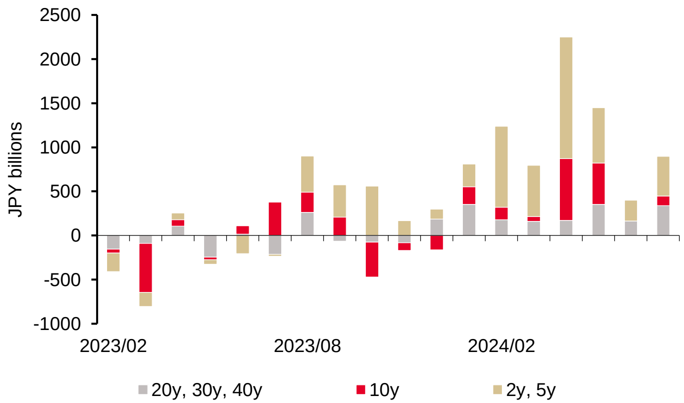
Source: SG Cross-Asset Research; JSDA data

Insurers heavily skew buys to long-end to fund liabilities



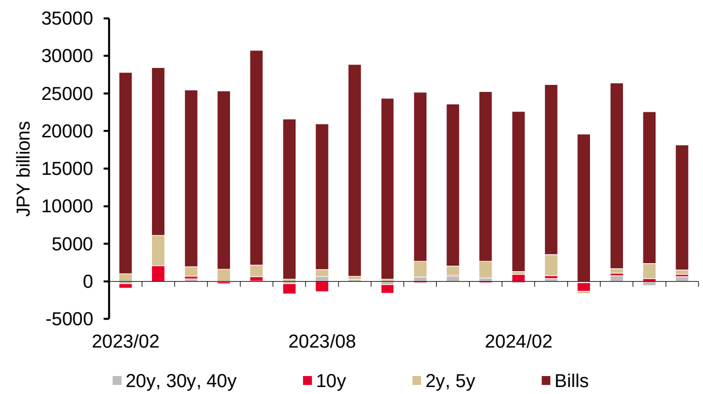
Source: SG Cross-Asset Research; JSDA data

Regional banks have been buying 2-5y as bank deposits have been rising



Source: SG Cross-Asset Research; JSDA data

Foreigner flow is often mixed but they are very large buyers of T-bills



Source: SG Cross-Asset Research; JSDA data

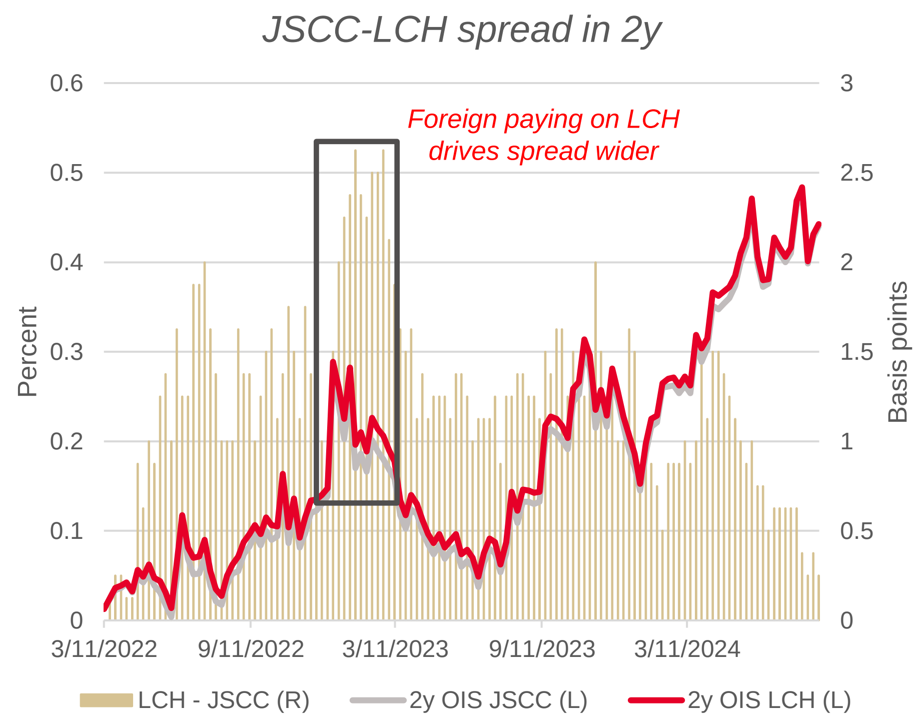
JPY SWAP RATES

- Almost all swap trades are JPY TONA OIS trades (Bloomberg: JYSO10 Curncy)
- There are two major clearers for JPY swaps and settlement prices often deviate due to supply/demand mismatches
- JSCC (Japan Securities Clearing Corp.) is mostly domestic funds, which tend to be receivers given local real money ALM needs. This has >70% of market share* (Bloomberg: MTRT or TPRA are common pricing codes e.g. JYSO10 MTRT Curncy)
- LCH SwapClear (London Clearing House) is dominated by overseas funds which tend to be better payers. ~30% market share* (Bloomberg: TPPA or ICLH are useful pricing codes)
- Rates settled on LCH tend to be higher than on JSCC (LCH-JSCC spread positive)
- Ability to clear through both and cost of posting double collateral means deviations are not easily arbitraged out

* ClarusFT: What's New in JPY Swaps in 2024? | (clarusft.com)

- <u>Following LCH-JSCC spread is a useful tool to add to the mosaic of positioning metrics</u>

- LCH-JSCC spreads widened sharply in 2022 as rates rose (Bloomberg: JYJLO2 Curncy)
- <u>Spreads widen as yields rise = LCH rates leading = overseas paying driving</u>
- Simple read-across of spread move. Direction of rates + direction of spread = flow sources
  - Rising rates + wider (tighter) LCH-JSCC spread = foreign (domestic) paying
  - Falling rates + wider (tighter) LCH-JSCC spread = foreign (domestic) receiving



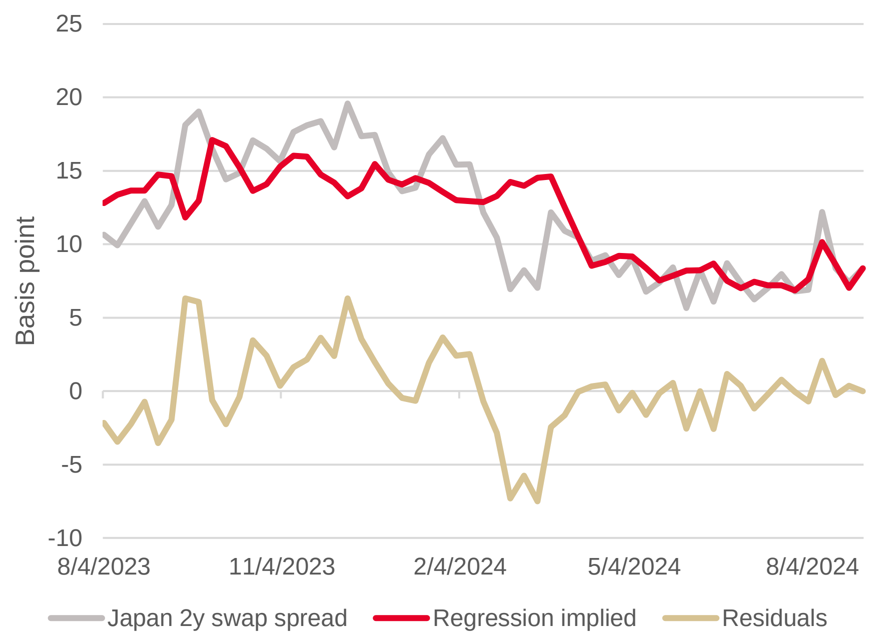
Source: SG Cross-Asset Research , Bloomberg

SWAP VS BONDS

Various factors drive the swap spread

- Swap spreads: Difference between swap rate and bond yield (Bloomberg: JYSST2 BLC Curncy)
- Drivers include:
- Spread between GC and TONA rates (a proxy for funding rates for bonds and swaps)
- QE stock + flow = changes in total BoJ % holdings of JGB supply + Rinban net flow in the sector
- Directional flows. Overseas paying fixed flows have contributed to bouts of spread widening
- ASW flows (either short bonds vs swap to capture higher yields, or pay swap vs bonds to hedge rate risk)

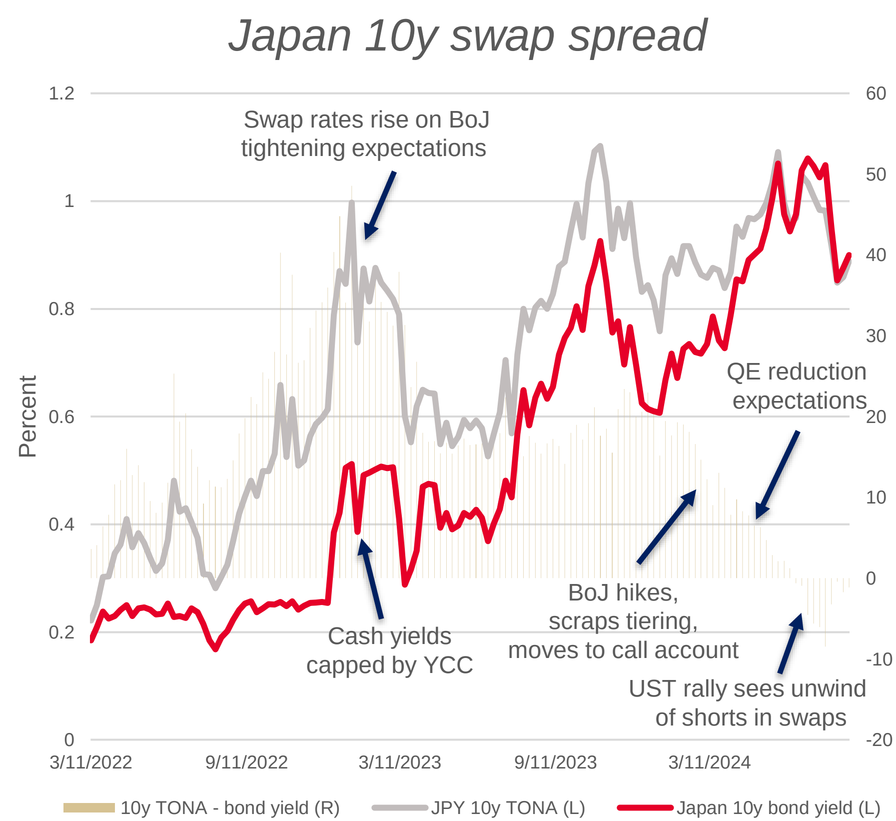
JPY 2y swap spread regression vs GC-TONA, QE stock + flow metrics, level of rates



Source: Bloomberg, SG Cross-Asset Research
Multiple R2 = 0.75
Regression = GC-TONA rolling average spread +JPY 2y swap rate + BoJ net flow in the sector + BoJ % holding of JGB market

SWAPS VS BONDS: RECENT HISTORY

- BoJ's Yield Curve Control pinned yields to 10y point while QE kept bond supply/demand tight (spreads wider)
- Hike expectations pushed swap rates higher. Speculators used swaps instead of bonds to avoid BoJ buys (spreads wider)
- Shift from rate tiering to call account in after first hike lifted GC rates vs TONA. Bonds are funded with in GC = bonds lagged (spreads tighter)
- Markets priced for further reductions in BoJ bond purchases, loosening tight supply demand (spreads tighter)
- UST rally/risk-off drives unwind of shorts expressed in swaps (spreads tighter)

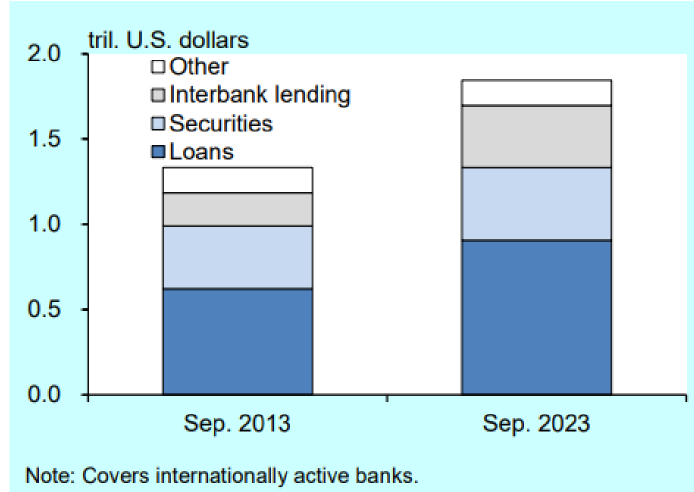


Sources: SG Cross Asset Research; Bloomberg

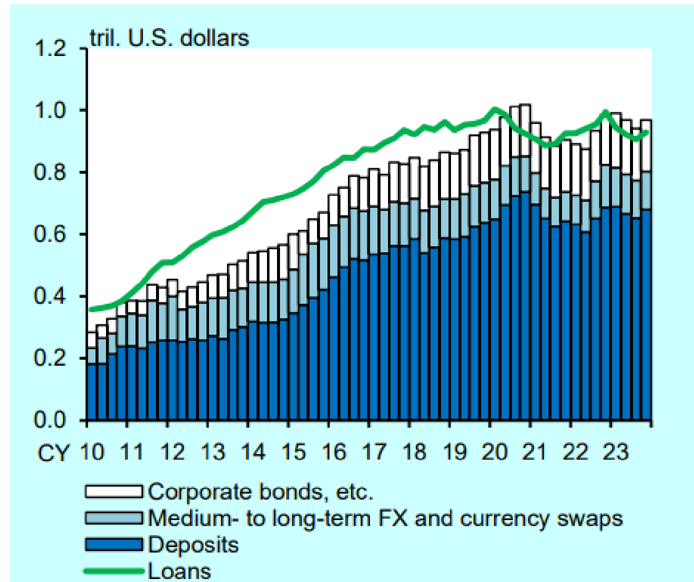
JAPAN: NEED THOSE DOLLARS

- Japanese banks have expanded loans and investment overseas over the past decade. Foreign currency balance sheets have surged as a result
- These banks do **not** have large, sticky USD deposits to fund it and rely on market-based funding (FX swap, corporate bonds) to bridge the gap
- Structural imbalance means USD carries a high premium to borrow vs JPY via funding markets
- <u>Overseas funds can lend USD for JPY to collect this premium and buy Japan bonds using XCCY basis or FX swap markets</u>
- These carry trades are popular. JPY proceeds are often typically invested by foreigners in shorter-dated tenors which have less duration risk
- <u>Over the past 12m overseas funds bought JPY22tn/month in Japan bills</u>*

Japan major banks' foreign currency assets



Source: BoJ Review: Foreign Currency Liquidity Risk Management at Japanese Major Banks: Efforts and Enhancement 01.pdf (fsa.go.jp)



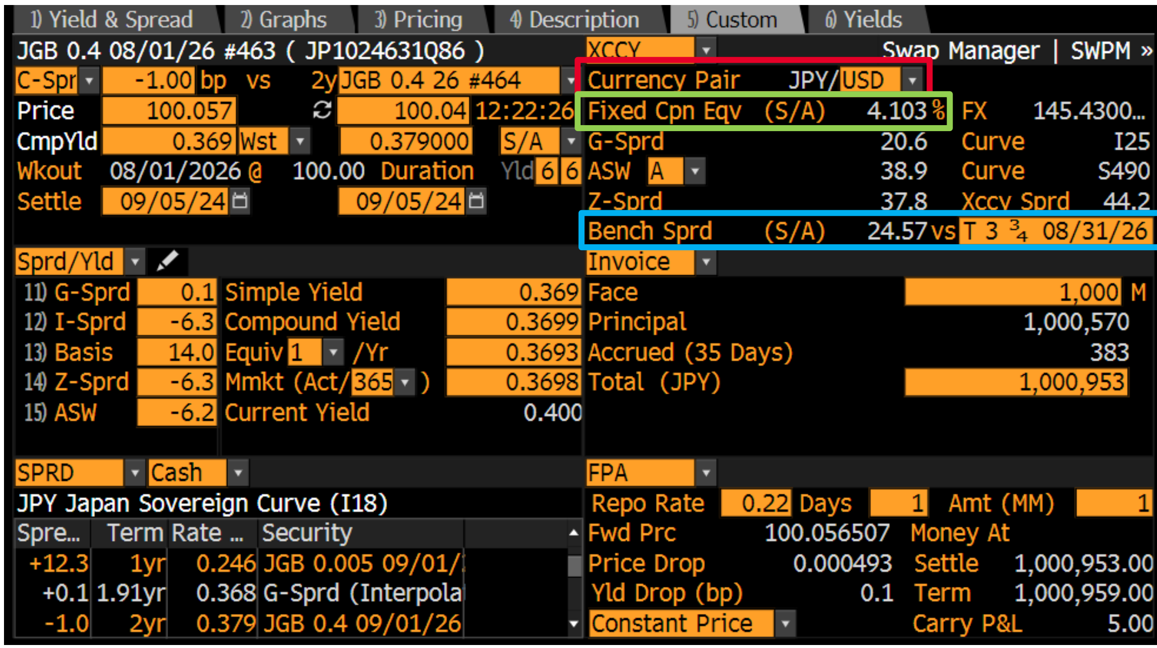
Source: BoJ Review: Foreign Currency Liquidity Risk Management at Japanese Major Banks: Efforts and Enhancement 01.pdf (fsa.go.jp)

* *JSDA* Bonds | Japan Securities Dealers Association (jsda.or.jp)

JAPAN: OVERSEAS HEDGED BUYERS

- Overseas buyers can harvest the USD premium (lend USD/borrow JPY & buy JP paper)

In practice: Load JGB ticker in Bloomberg. Hit YAS XCCY. Set currency pair (red) fixed coupon equivalent (green) and see all in pick-up vs comparable local bond (blue)



* Source: Bloomberg

Bigger picture: SG Research publishes this bi-weekly DM/EM ASW monitor

USD Investor

pick up over US Treasuries (bp)

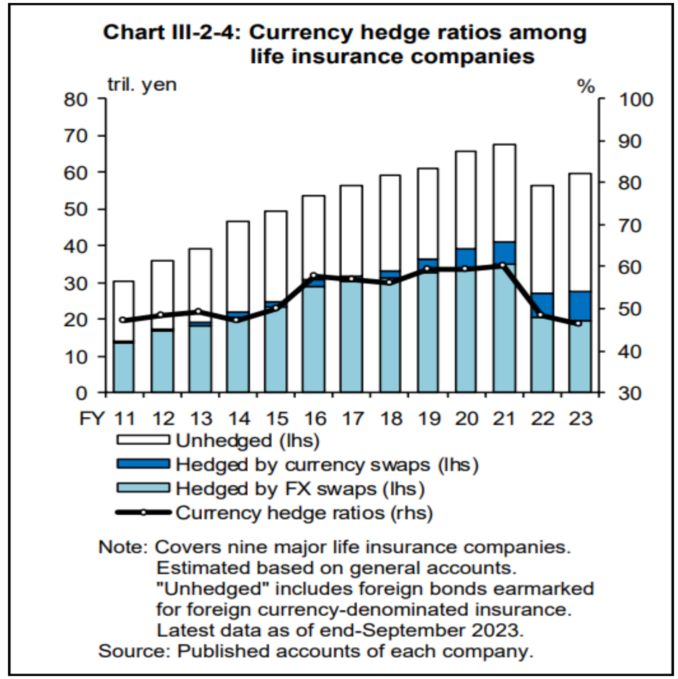
	1Y	2Y	3Y	4Y	5Y	7Y	10Y	15Y
JPY	26	23	21	18	24	21	35	47
CAD	19	-2	-1	-10	-9	-17	-29	
CHF	18	-2	-12	-15	-18	-27	-37	-39
NOK	12		-29	-36	-29	-28	-28	
SEK		-40			-43	-54	-61	-70
AUD	20	-24	-30	-35	-35	-42	-53	-51
US	0	0	0	0	0	0	0	0
GBP	-7	-45	-22	-8	-15	-14	0	0
germany	-4	-23	-32	-32	-30	-41	-40	-37
france	16	0	10	12	21	18	37	34
spain	14	10	14	11	22	30	49	73
italy	19	27	39	41	64	85	107	135

* Source: SG Research, Bloomberg. Full note available here

JAPAN: CREDITORS TO THE WORLD

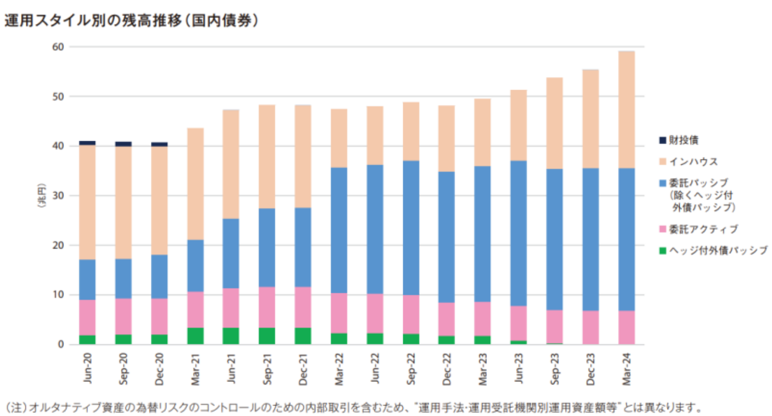
- Japan is the largest net foreign asset holder at $3 trillion*
- Low rates makes FX hedging expensive. Weak JPY + expensive hedges = lifers and pension funds have cut FX hedges in recent years
- Typically, domestic investors treat FX-hedged and domestic bonds as the same. Less FX hedged bonds = more domestic bond demand

Lifer FX-hedges lowest since at least 2011



* Source: Japan financial system report April 2024.

Japan's largest pensions fund (GPIF) shrunk FX-hedged bonds (green) while adding total inhouse actively managed domestic bonds holdings (orange)



* Source: GPIF disclosures March 2024

- After strong JPY rally in July/August 2024, participants worried Japanese would repatriate foreign bond holdings (or raise hedge ratios)
- This was not the case. The response: they bought over JPY10tn of foreign assets over the following six weeks.

* Source: Japan net external assets hit record high in 2023, remains world's top creditor | Reuters

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